treat them as unknown for the time being because we intend to show that Snell's laws also can be derived by applying field boundary conditions at $z = 0$. The total electric field in medium 1 is the sum of the incident and reflected electric fields: $\widetilde{\mathbf{E}}_\perp^1 = \widetilde{\mathbf{E}}_\perp^i + \widetilde{\mathbf{E}}_\perp^r$; a similar statement holds true for the total magnetic field in medium 1: $\widetilde{\mathbf{H}}_\perp^1 = \widetilde{\mathbf{H}}_\perp^i + \widetilde{\mathbf{H}}_\perp^r$. Boundary conditions state that the tangential components of $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ each must be continuous across the boundary between the two media. Field components tangential to the boundary extend along $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$. Since the electric fields in media 1 and 2 have $\hat{\mathbf{y}}$ components only, the boundary condition for $\widetilde{\mathbf{E}}$ is

$$(\widetilde{E}_{\perp y}^i + \widetilde{E}_{\perp y}^r)\Big|_{z=0} = \widetilde{E}_{\perp y}^t\Big|_{z=0}. \tag{8.50}$$

Upon using Eqs. (8.48a), (8.49a), and (8.49c) in Eq. (8.50) and then setting $z = 0$, we have

$$E_{\perp 0}^i e^{-jk_1 x \sin\theta_i} + E_{\perp 0}^r e^{-jk_1 x \sin\theta_r} = E_{\perp 0}^t e^{-jk_2 x \sin\theta_t}. \tag{8.51}$$

Since the magnetic fields in media 1 and 2 have no $\hat{\mathbf{y}}$ components, the boundary condition for $\widetilde{\mathbf{H}}$ is

$$(\widetilde{H}_{\perp x}^i + \widetilde{H}_{\perp x}^r)\Big|_{z=0} = \widetilde{H}_{\perp x}^t\Big|_{z=0}, \tag{8.52}$$

or

$$-\frac{E_{\perp 0}^i}{\eta_1}\cos\theta_i\, e^{-jk_1 x \sin\theta_i} + \frac{E_{\perp 0}^r}{\eta_1}\cos\theta_r\, e^{-jk_1 x \sin\theta_r} = -\frac{E_{\perp 0}^t}{\eta_2}\cos\theta_t\, e^{-jk_2 x \sin\theta_t}. \tag{8.53}$$

To satisfy Eqs. (8.51) and (8.53) for all possible values of x (i.e., all along the boundary), it follows that the arguments of all three exponentials must be equal. That is,

$$k_1 \sin\theta_i = k_1 \sin\theta_r = k_2 \sin\theta_t, \tag{8.54}$$

which is known as the ***phase-matching condition***. The first equality in Eq. (8.54) leads to

$$\theta_r = \theta_i, \qquad \textbf{(Snell's law of reflection)} \tag{8.55}$$

while the second equality leads to

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{k_1}{k_2} = \frac{\omega\sqrt{\mu_1\varepsilon_1}}{\omega\sqrt{\mu_2\varepsilon_2}} = \frac{n_1}{n_2}. \tag{8.56}$$

(Snell's law of refraction)

The results expressed by Eqs. (8.55) and (8.56) are identical with those derived previously in Section 8-2 through consideration of the ray path traversed by the incident, reflected, and transmitted wavefronts.

In view of Eq. (8.54), the boundary conditions given by Eqs. (8.51) and (8.53) reduce to

$$E_{\perp 0}^i + E_{\perp 0}^r = E_{\perp 0}^t, \tag{8.57a}$$

$$\frac{\cos\theta_i}{\eta_1}(-E_{\perp 0}^i + E_{\perp 0}^r) = -\frac{\cos\theta_t}{\eta_2} E_{\perp 0}^t. \tag{8.57b}$$

These two equations can be solved simultaneously to yield the following expressions for the reflection and transmission coefficients in the perpendicular polarization case:

$$\Gamma_\perp = \frac{E_{\perp 0}^r}{E_{\perp 0}^i} = \frac{\eta_2\cos\theta_i - \eta_1\cos\theta_t}{\eta_2\cos\theta_i + \eta_1\cos\theta_t}, \tag{8.58a}$$

$$\tau_\perp = \frac{E_{\perp 0}^t}{E_{\perp 0}^i} = \frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_i + \eta_1\cos\theta_t}. \tag{8.58b}$$

These two coefficients are known formally as the ***Fresnel reflection and transmission coefficients for perpendicular polarization*** and are related by

$$\tau_\perp = 1 + \Gamma_\perp. \tag{8.59}$$

If medium 2 is a perfect conductor ($\eta_2 = 0$), Eqs. (8.58a and b) reduce to $\Gamma_\perp = -1$ and $\tau_\perp = 0$, respectively, which means that the incident wave is totally reflected by the conducting medium.

For nonmagnetic dielectrics with $\mu_1 = \mu_2 = \mu_0$ and with the help of Eq. (8.56), the expression for $\Gamma_\perp$ can be written as

$$\Gamma_\perp = \frac{\cos\theta_i - \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i}}{\cos\theta_i + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i}} \quad (\text{for } \mu_1 = \mu_2). \tag{8.60}$$

Since $(\varepsilon_2/\varepsilon_1) = (n_2/n_1)^2$, this expression also can be written in terms of the indices of refraction n_1 and n_2.

暂时将它们视为未知，因为我们打算证明斯涅耳定律也可以通过在 z = 0 处应用场边界条件来导出。介质 1 中的总电场是入射电场和反射电场的总和：$\widetilde{\mathrm{E}}_\perp^1 = \widetilde{\mathrm{E}}_\perp^i + \widetilde{\mathrm{E}}_\perp^r$；类似的说法也适用于介质 1 中的总磁场：$\widetilde{\mathrm{H}}_\perp^1 = \widetilde{\mathrm{H}}_\perp^i + \widetilde{\mathrm{H}}_\perp^r$。边界条件规定 $\widetilde{\mathrm{E}}$ 和 $\widetilde{\mathrm{H}}$ 的切向分量必须在两种介质之间的边界上连续。与边界相切的场分量沿 $\hat{\mathrm{x}}$ 和 $\hat{\mathrm{y}}$ 延伸。由于介质 1 和 2 中的电场仅具有 $\hat{\mathrm{y}}$ 分量，因此 $\widetilde{\mathrm{E}}$ 的边界条件为

$$(\widetilde{E}_{\perp y}^i + \widetilde{E}_{\perp y}^r)\Big|_{z=0} = \widetilde{E}_{\perp y}^t\Big|_{z=0}. \tag{8.50}$$

使用方程式后。式（8.48a）、（8.49a）和（8.49c）(8.50) 然后设置 z = 0，我们有

$$E_{\perp 0}^i e^{-jk_1 x \sin\theta_i} + E_{\perp 0}^r e^{-jk_1 x \sin\theta_r} = E_{\perp 0}^t e^{-jk_2 x \sin\theta_t}. \tag{8.51}$$

由于介质 1 和 2 中的磁场没有 $\hat{\mathrm{y}}$ 分量，因此 $\widetilde{\mathrm{H}}$ 的边界条件为

$$(\widetilde{H}_{\perp x}^i + \widetilde{H}_{\perp x}^r)\Big|_{z=0} = \widetilde{H}_{\perp x}^t\Big|_{z=0}, \tag{8.52}$$

或者

$$-\frac{E_{\perp 0}^i}{\eta_1}\cos\theta_i\, e^{-jk_1 x \sin\theta_i} + \frac{E_{\perp 0}^r}{\eta_1}\cos\theta_r\, e^{-jk_1 x \sin\theta_r} = -\frac{E_{\perp 0}^t}{\eta_2}\cos\theta_t\, e^{-jk_2 x \sin\theta_t}. \tag{8.53}$$

满足方程。(8.51) 和 (8.53) 对于 x 的所有可能值（即，沿着边界），所有三个指数的参数必须相等。那是，

$$k_1 \sin\theta_i = k_1 \sin\theta_r = k_2 \sin\theta_t, \tag{8.54}$$

这称为相位匹配条件。方程中的第一个等式。(8.54) 导致

$$\theta_r = \theta_i, \qquad \textbf{(Snell's law of reflection)} \tag{8.55}$$

而第二个等式导致

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{k_1}{k_2} = \frac{\omega\sqrt{\mu_1\varepsilon_1}}{\omega\sqrt{\mu_2\varepsilon_2}} = \frac{n_1}{n_2}. \tag{8.56}$$

(Snell's law of refraction)

结果由方程式表示。（8.55）和（8.56）与前面在第 8-2 节中通过考虑入射、反射和透射波前穿过的光线路径而推导出来的相同。

鉴于方程。（8.54），由方程给出的边界条件。(8.51) 和 (8.53) 减少为

$$E_{\perp 0}^i + E_{\perp 0}^r = E_{\perp 0}^t, \tag{8.57a}$$

$$\frac{\cos\theta_i}{\eta_1}(-E_{\perp 0}^i + E_{\perp 0}^r) = -\frac{\cos\theta_t}{\eta_2} E_{\perp 0}^t. \tag{8.57b}$$

可以同时求解这两个方程，得出垂直偏振情况下反射和透射系数的以下表达式：

$$\Gamma_\perp = \frac{E_{\perp 0}^r}{E_{\perp 0}^i} = \frac{\eta_2\cos\theta_i - \eta_1\cos\theta_t}{\eta_2\cos\theta_i + \eta_1\cos\theta_t}, \tag{8.58a}$$

$$\tau_\perp = \frac{E_{\perp 0}^t}{E_{\perp 0}^i} = \frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_i + \eta_1\cos\theta_t}. \tag{8.58b}$$

这两个系数的正式名称为垂直偏振的菲涅尔反射和透射系数，其关系式为

$$\tau_\perp = 1 + \Gamma_\perp. \tag{8.59}$$

如果介质 2 是理想导体 ($\eta_2 = 0$)，则等式：(8.58a和b)分别减少到$\Gamma_\perp = -1$和$\tau_\perp = 0$，这意味着入射波被导电介质完全反射。

对于具有 $\mu_1 = \mu_2 = \mu_0$ 的非磁性电介质，并借助方程式。(8.56)，$\Gamma_\perp$ 的表达式可以写为

$$\Gamma_\perp = \frac{\cos\theta_i - \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i}}{\cos\theta_i + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_i}} \quad (\text{for } \mu_1 = \mu_2). \tag{8.60}$$

由于 $(\varepsilon_2/\varepsilon_1) = (n_2/n_1)^2$，该表达式也可以用折射率 n_1 和 n_2 来编写。

Example 8-6: Wave Incident Obliquely on a Soil Surface

Using the coordinate system of Fig. 8-15, a plane wave radiated by a distant antenna is incident in air upon a plane soil surface located at $z = 0$. The electric field of the incident wave is given by

$$\mathbf{E}^{\mathrm{i}} = \hat{\mathbf{y}}100\cos(\omega t - \pi x - 1.73\pi z) \quad \text{(V/m)}, \tag{8.61}$$

and the soil medium may be assumed to be a lossless dielectric with a relative permittivity of 4.

(a) Determine k_1, k_2, and the incidence angle θ_{i}.
(b) Obtain expressions for the total electric fields in air and in the soil.
(c) Determine the average power density carried by the wave traveling in soil.

Solution: (a) We begin by converting Eq. (8.61) into phasor form, akin to the expression given by Eq. (8.46a):

$$\widetilde{\mathbf{E}}^{\mathrm{i}} = \hat{\mathbf{y}}100e^{-j\pi x - j1.73\pi z} = \hat{\mathbf{y}}100e^{-jk_1 x_{\mathrm{i}}} \quad \text{(V/m)}, \tag{8.62}$$

where x_{i} is the axis along which the wave is traveling and

$$k_1 x_{\mathrm{i}} = \pi x + 1.73\pi z. \tag{8.63}$$

Using Eq. (8.47a), we have

$$k_1 x_{\mathrm{i}} = k_1 x \sin\theta_{\mathrm{i}} + k_1 z \cos\theta_{\mathrm{i}}. \tag{8.64}$$

Hence,

$$k_1 \sin\theta_{\mathrm{i}} = \pi,$$
$$k_1 \cos\theta_{\mathrm{i}} = 1.73\pi,$$

which together give

$$k_1 = \sqrt{\pi^2 + (1.73\pi)^2} = 2\pi \quad \text{(rad/m)},$$
$$\theta_{\mathrm{i}} = \tan^{-1}\left(\frac{\pi}{1.73\pi}\right) = 30^\circ.$$

The wavelength in medium 1 (air) is

$$\lambda_1 = \frac{2\pi}{k_1} = 1 \text{ m},$$

and the wavelength in medium 2 (soil) is

$$\lambda_2 = \frac{\lambda_1}{\sqrt{\varepsilon_{r_2}}} = \frac{1}{\sqrt{4}} = 0.5 \text{ m}.$$

The corresponding wavenumber in medium 2 is

$$k_2 = \frac{2\pi}{\lambda_2} = 4\pi \quad \text{(rad/m)}.$$

Since $\widetilde{\mathbf{E}}^{\mathrm{i}}$ is along $\hat{\mathbf{y}}$, it is perpendicularly polarized ($\hat{\mathbf{y}}$ is perpendicular to the plane of incidence containing the surface normal $\hat{\mathbf{z}}$ and the propagation direction $\hat{\mathbf{x}}_{\mathrm{i}}$).

(b) Given that $\theta_{\mathrm{i}} = 30^\circ$, the transmission angle θ_{t} is obtained with the help of Eq. (8.56):

$$\sin\theta_{\mathrm{t}} = \frac{k_1}{k_2}\sin\theta_{\mathrm{i}} = \frac{2\pi}{4\pi}\sin 30^\circ = 0.25$$

or

$$\theta_{\mathrm{t}} = 14.5^\circ.$$

With $\varepsilon_1 = \varepsilon_0$ and $\varepsilon_2 = \varepsilon_{r_2}\varepsilon_0 = 4\varepsilon_0$, the reflection and transmission coefficients for perpendicular polarization are determined with the help of Eqs. (8.59) and (8.60),

$$\Gamma_\perp = \frac{\cos\theta_{\mathrm{i}} - \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_{\mathrm{i}}}}{\cos\theta_{\mathrm{i}} + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_{\mathrm{i}}}} = -0.38,$$

$$\tau_\perp = 1 + \Gamma_\perp = 0.62.$$

Using Eqs. (8.48a) and (8.49a) with $E^{\mathrm{i}}_{\perp 0} = 100$ V/m and $\theta_{\mathrm{i}} = \theta_{\mathrm{r}}$, the total electric field in medium 1 is

$$\begin{aligned}\widetilde{\mathbf{E}}^1_\perp &= \widetilde{\mathbf{E}}^{\mathrm{i}}_\perp + \widetilde{\mathbf{E}}^{\mathrm{r}}_\perp \\ &= \hat{\mathbf{y}}E^{\mathrm{i}}_{\perp 0}e^{-jk_1(x\sin\theta_{\mathrm{i}} + z\cos\theta_{\mathrm{i}})} + \hat{\mathbf{y}}\Gamma E^{\mathrm{i}}_{\perp 0}e^{-jk_1(x\sin\theta_{\mathrm{i}} - z\cos\theta_{\mathrm{i}})} \\ &= \hat{\mathbf{y}}100e^{-j(\pi x + 1.73\pi z)} - \hat{\mathbf{y}}38e^{-j(\pi x - 1.73\pi z)},\end{aligned}$$

and the corresponding instantaneous electric field in medium 1 is

$$\begin{aligned}\mathbf{E}^1_\perp(x,z,t) &= \mathfrak{Re}\left[\widetilde{\mathbf{E}}^1_\perp e^{j\omega t}\right] \\ &= \hat{\mathbf{y}}[100\cos(\omega t - \pi x - 1.73\pi z) \\ &\quad - 38\cos(\omega t - \pi x + 1.73\pi z)] \quad \text{(V/m)}.\end{aligned}$$

In medium 2, using Eq. (8.49c) with $E^{\mathrm{t}}_{\perp 0} = \tau_\perp E^{\mathrm{i}}_{\perp 0}$ gives

$$\widetilde{\mathbf{E}}^{\mathrm{t}}_\perp = \hat{\mathbf{y}}\tau E^{\mathrm{i}}_{\perp 0}e^{-jk_2(x\sin\theta_{\mathrm{i}} + z\cos\theta_{\mathrm{i}})} = \hat{\mathbf{y}}62e^{-j(\pi x + 3.87\pi z)}$$

and, correspondingly,

$$\begin{aligned}\mathbf{E}^{\mathrm{t}}_\perp(x,z,t) &= \mathfrak{Re}\left[\widetilde{\mathbf{E}}^{\mathrm{t}}_\perp e^{j\omega t}\right] \\ &= \hat{\mathbf{y}}62\cos(\omega t - \pi x - 3.87\pi z) \quad \text{(V/m)}.\end{aligned}$$

(c) In medium 2, $\eta_2 = \eta_0/\sqrt{\varepsilon_{r_2}} \approx 120\pi/\sqrt{4} = 60\pi\ (\Omega)$, and the average power density carried by the wave is

$$S^{\mathrm{t}}_{\mathrm{av}} = \frac{|E^{\mathrm{t}}_{\perp 0}|^2}{2\eta_2} = \frac{(62)^2}{2\times 60\pi} = 10.2 \quad \text{(W/m}^2\text{)}.$$

例8-6：波浪倾斜入射到土壤表面

使用图 8-15 的坐标系，由远处天线辐射的平面波在空气中入射到位于 z = 0 处的平面土壤表面上。入射波的电场由下式给出

$$\mathbf{E}^{\mathrm{i}} = \hat{\mathbf{y}}100\cos(\omega t - \pi x - 1.73\pi z) \quad \text{(V/m)}, \tag{8.61}$$

土壤介质可假设为相对介电常数为 4 的无损电介质。

(a) 确定 k_1、k_2 和入射角 θ_{i}。(b) 获得空气和土壤中总电场的表达式。(c) 确定在土壤中传播的波所携带的平均功率密度。

解决方案：(a) 我们首先将方程转换为：（8.61）转化为相量形式，类似于方程给出的表达式。（8.46a）：

$$\widetilde{\mathbf{E}}^{\mathrm{i}} = \hat{\mathbf{y}}100e^{-j\pi x - j1.73\pi z} = \hat{\mathbf{y}}100e^{-jk_1 x_{\mathrm{i}}} \quad \text{(V/m)}, \tag{8.62}$$

其中 x_{i} 是波传播的轴，

$$k_1 x_{\mathrm{i}} = \pi x + 1.73\pi z. \tag{8.63}$$

使用方程式（8.47a），我们有

$$k_1 x_{\mathrm{i}} = k_1 x \sin\theta_{\mathrm{i}} + k_1 z \cos\theta_{\mathrm{i}}. \tag{8.64}$$

因此，

$$k_1 \sin\theta_{\mathrm{i}} = \pi,$$
$$k_1 \cos\theta_{\mathrm{i}} = 1.73\pi,$$

一起给予

$$k_1 = \sqrt{\pi^2 + (1.73\pi)^2} = 2\pi \quad \text{(rad/m)},$$
$$\theta_{\mathrm{i}} = \tan^{-1}\left(\frac{\pi}{1.73\pi}\right) = 30^\circ.$$

介质 1（空气）中的波长为

$$\lambda_1 = \frac{2\pi}{k_1} = 1 \text{ m},$$

介质 2（土壤）中的波长为

$$\lambda_2 = \frac{\lambda_1}{\sqrt{\varepsilon_{r_2}}} = \frac{1}{\sqrt{4}} = 0.5 \text{ m}.$$

介质2中对应的波数为

$$k_2 = \frac{2\pi}{\lambda_2} = 4\pi \quad \text{(rad/m)}.$$

由于 $\widetilde{\mathbf{E}}^{\mathrm{i}}$ 沿 $\hat{\mathbf{y}}$，因此它是垂直偏振的（$\hat{\mathbf{y}}$ 垂直于包含表面法线 $\hat{\mathbf{z}}$ 和传播方向 $\hat{\mathbf{x}}_{\mathrm{i}}$ 的入射平面）。

(b) 给定$\theta_{\mathrm{i}} = 30^\circ$，则借助方程(1)获得传输角$\theta_{\mathrm{t}}$。（8.56）：

$$\sin\theta_{\mathrm{t}} = \frac{k_1}{k_2}\sin\theta_{\mathrm{i}} = \frac{2\pi}{4\pi}\sin 30^\circ = 0.25$$

or

$$\theta_{\mathrm{t}} = 14.5^\circ.$$

对于 $\varepsilon_1 = \varepsilon_0$ 和 $\varepsilon_2 = \varepsilon_{r_2}\varepsilon_0 = 4\varepsilon_0$，垂直偏振的反射和透射系数可借助方程 (1) 确定。（8.59）和（8.60），

$$\Gamma_\perp = \frac{\cos\theta_{\mathrm{i}} - \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_{\mathrm{i}}}}{\cos\theta_{\mathrm{i}} + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_{\mathrm{i}}}} = -0.38,$$

$$\tau_\perp = 1 + \Gamma_\perp = 0.62.$$

使用方程式。(8.48a) 和 (8.49a) 当 $E^{\mathrm{i}}_{\perp 0} = 100$ V/m 和 $\theta_{\mathrm{i}} = \theta_{\mathrm{r}}$ 时，介质 1 中的总电场为

$$\begin{aligned}\widetilde{\mathbf{E}}^1_\perp &= \widetilde{\mathbf{E}}^{\mathrm{i}}_\perp + \widetilde{\mathbf{E}}^{\mathrm{r}}_\perp \\ &= \hat{\mathbf{y}}E^{\mathrm{i}}_{\perp 0}e^{-jk_1(x\sin\theta_{\mathrm{i}} + z\cos\theta_{\mathrm{i}})} + \hat{\mathbf{y}}\Gamma E^{\mathrm{i}}_{\perp 0}e^{-jk_1(x\sin\theta_{\mathrm{i}} - z\cos\theta_{\mathrm{i}})} \\ &= \hat{\mathbf{y}}100e^{-j(\pi x + 1.73\pi z)} - \hat{\mathbf{y}}38e^{-j(\pi x - 1.73\pi z)},\end{aligned}$$

介质1中相应的瞬时电场为

$$\begin{aligned}\mathbf{E}^1_\perp(x,z,t) &= \mathfrak{Re}\left[\widetilde{\mathbf{E}}^1_\perp e^{j\omega t}\right] \\ &= \hat{\mathbf{y}}[100\cos(\omega t - \pi x - 1.73\pi z) \\ &\quad - 38\cos(\omega t - \pi x + 1.73\pi z)] \quad \text{(V/m)}.\end{aligned}$$

在介质 2 中，使用等式：(8.49c) 与 $\mathbf{E}^{\mathrm{t}}_{\perp 0} = \tau_\perp \mathbf{E}^{\mathrm{i}}_{\perp 0}$ 给出

$$\widetilde{\mathbf{E}}^{\mathrm{t}}_\perp = \hat{\mathbf{y}}\tau E^{\mathrm{i}}_{\perp 0}e^{-jk_2(x\sin\theta_{\mathrm{i}} + z\cos\theta_{\mathrm{i}})} = \hat{\mathbf{y}}62e^{-j(\pi x + 3.87\pi z)}$$

并且，相应地，

$$\begin{aligned}\mathbf{E}^{\mathrm{t}}_\perp(x,z,t) &= \mathfrak{Re}\left[\widetilde{\mathbf{E}}^{\mathrm{t}}_\perp e^{j\omega t}\right] \\ &= \hat{\mathbf{y}}62\cos(\omega t - \pi x - 3.87\pi z) \quad \text{(V/m)}.\end{aligned}$$

(c) 在介质 2 中，$\eta_2 = \eta_0/\sqrt{\varepsilon_{r_2}} \approx 120\pi/\sqrt{4} = 60\pi\ (\Omega)$，波携带的平均功率密度为

$$S^{\mathrm{t}}_{\mathrm{av}} = \frac{|E^{\mathrm{t}}_{\perp 0}|^2}{2\eta_2} = \frac{(62)^2}{2\times 60\pi} = 10.2 \quad \text{(W/m}^2\text{)}.$$

Technology Brief 16: Bar-Code Readers

A ***bar code*** consists of a sequence of parallel bars of certain widths, usually printed in black against a white background, configured to represent a particular ***binary code*** of information about a product and its manufacturer. ***Laser scanners*** can read the code and transfer the information to a computer, a cash register, or a display screen. For both stationary scanners built into checkout counters at grocery stores and handheld units that can be pointed at the bar-coded object like a gun, the basic operation of a bar-code reader is the same.

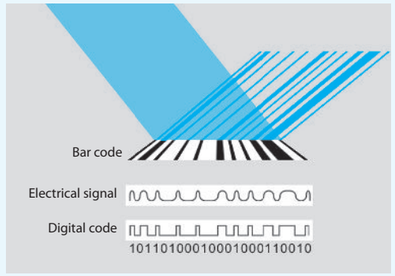


Figure TF16-2 Bar code contained in reflected laser beam.

Basic Operation

The scanner uses a laser beam of light pointed at a multifaceted ***rotating mirror***, spinning at a high speed on the order of 6,000 revolutions per minute (**Fig. TF16-1**). The rotating mirror creates a ***fan beam*** to illuminate the bar code on the object. Moreover, by exposing the laser light to its many facets, it deflects the beam into many different directions, allowing the object to be scanned over a wide range of positions and orientations. The goal is to have one of those directions be such that the beam reflected by the bar code ends up traveling in the direction of, and captured by, the light detector (***sensor***), which then reads the coded sequence (white bars reflect laser light and black ones do not) and converts it into a binary sequence of ones and zeros (**Fig. TF16-2**). To eliminate interference by ambient light, a ***glass filter*** is used as shown in **Fig. TF16-1** to block out all light except for a narrow wavelength band centered at the wavelength of the laser light.

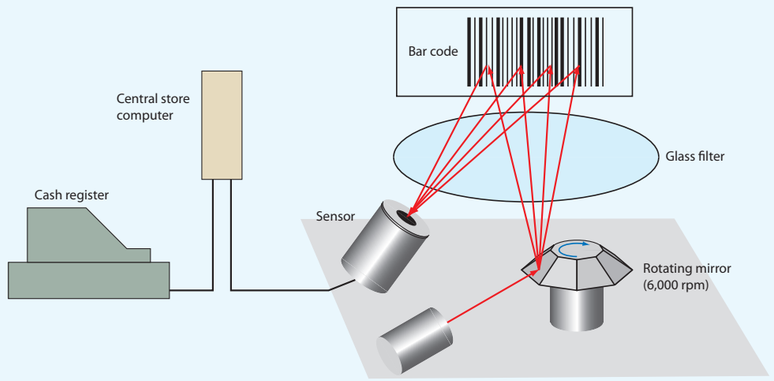


Figure TF16-1 Elements of a bar-code reader.

技术简介 16：条码读取器

条形码由一系列具有一定宽度的平行条组成，通常以白色背景上的黑色印刷，配置为表示有关产品及其制造商的信息的特定二进制代码。激光扫描仪可以读取代码并将信息传输到计算机、收银机或显示屏。对于内置于杂货店收银台的固定扫描仪和可以像枪一样指向条形码物体的手持设备，条形码阅读器的基本操作是相同的。

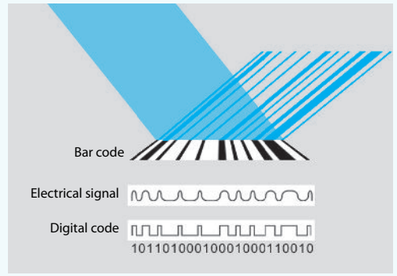


图 TF16-2 反射激光束中包含的条形码。

基本操作

扫描仪使用激光束指向多面旋转镜，该镜以每分钟 6,000 转的速度高速旋转（图 TF16-1）。旋转镜产生扇形光束来照亮物体上的条形码。此外，通过将激光照射到其多个面，它可以将光束偏转到许多不同的方向，从而可以在广泛的位置和方向上扫描物体。目标是使这些方向之一使光束反射

条形码最终沿光检测器（传感器）的方向行进并被光检测器（传感器）捕获，然后读取编码序列（白条反射激光，黑条不反射激光）并将其转换为 1 和 0 的二进制序列（图 TF16-2）。为了消除环境光的干扰，如图TF16-1所示，使用玻璃滤光片来阻挡除以激光波长为中心的窄波长带之外的所有光。

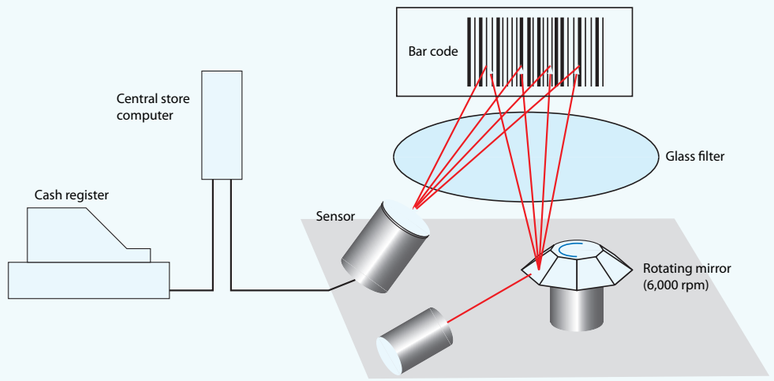


图 TF16-1 条形码阅读器的元件。

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8-4.2 Parallel Polarization

If we interchange the roles played by **E** and **H** in the perpendicular polarization scenario covered in the preceding subsection, while keeping in mind the requirement that $\mathbf{E}\times\mathbf{H}$ must point in the direction of propagation for each of the incident, reflected, and transmitted waves, we end up with the parallel polarization scenario shown in **Fig. 8-16**. Now the electric fields lie in the plane of incidence, while the associated magnetic fields are perpendicular to the plane of incidence. With reference to the directions indicated in **Fig. 8-16**, the fields of the incident, reflected, and transmitted waves are given by

Incident Wave

$$\widetilde{\mathbf{E}}_{\parallel}^{\mathrm{i}} = \hat{\mathbf{y}}_{\mathrm{i}} E_{\parallel 0}^{\mathrm{i}} e^{-jk_1 x_{\mathrm{i}}} = (\hat{\mathbf{x}}\cos\theta_{\mathrm{i}} - \hat{\mathbf{z}}\sin\theta_{\mathrm{i}}) E_{\parallel 0}^{\mathrm{i}} e^{-jk_1(x\sin\theta_{\mathrm{i}} + z\cos\theta_{\mathrm{i}})}, \tag{8.65a}$$

$$\widetilde{\mathbf{H}}_{\parallel}^{\mathrm{i}} = \hat{\mathbf{y}}\,\frac{E_{\parallel 0}^{\mathrm{i}}}{\eta_1} e^{-jk_1 x_{\mathrm{i}}} = \hat{\mathbf{y}}\,\frac{E_{\parallel 0}^{\mathrm{i}}}{\eta_1} e^{-jk_1(x\sin\theta_{\mathrm{i}} + z\cos\theta_{\mathrm{i}})}, \tag{8.65b}$$

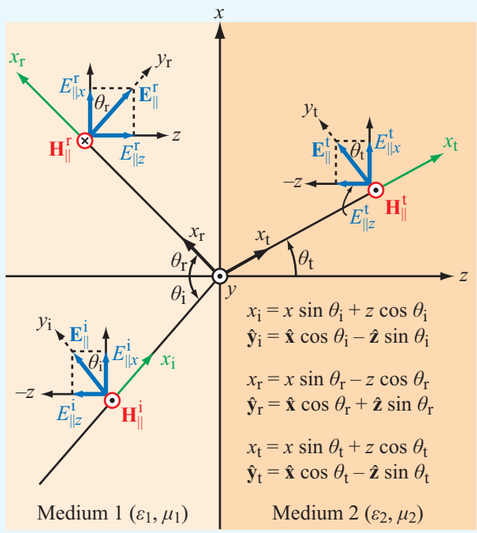


Figure 8-16 Parallel-polarized plane wave incident at an angle θ_{i} upon a planar boundary.

Reflected Wave

$$\widetilde{\mathbf{E}}_{\parallel}^{\mathrm{r}} = \hat{\mathbf{y}}_{\mathrm{r}} E_{\parallel 0}^{\mathrm{r}} e^{-jk_1 x_{\mathrm{r}}} = (\hat{\mathbf{x}}\cos\theta_{\mathrm{r}} + \hat{\mathbf{z}}\sin\theta_{\mathrm{r}}) E_{\parallel 0}^{\mathrm{r}} e^{-jk_1(x\sin\theta_{\mathrm{r}} - z\cos\theta_{\mathrm{r}})}, \tag{8.65c}$$

$$\widetilde{\mathbf{H}}_{\parallel}^{\mathrm{r}} = -\hat{\mathbf{y}}\,\frac{E_{\parallel 0}^{\mathrm{r}}}{\eta_1} e^{-jk_1 x_{\mathrm{r}}} = -\hat{\mathbf{y}}\,\frac{E_{\parallel 0}^{\mathrm{r}}}{\eta_1} e^{-jk_1(x\sin\theta_{\mathrm{r}} - z\cos\theta_{\mathrm{r}})}, \tag{8.65d}$$

Transmitted Wave

$$\widetilde{\mathbf{E}}_{\parallel}^{\mathrm{t}} = \hat{\mathbf{y}}_{\mathrm{t}} E_{\parallel 0}^{\mathrm{t}} e^{-jk_2 x_{\mathrm{t}}} = (\hat{\mathbf{x}}\cos\theta_{\mathrm{t}} - \hat{\mathbf{z}}\sin\theta_{\mathrm{t}}) E_{\parallel 0}^{\mathrm{t}} e^{-jk_2(x\sin\theta_{\mathrm{t}} + z\cos\theta_{\mathrm{t}})}, \tag{8.65e}$$

$$\widetilde{\mathbf{H}}_{\parallel}^{\mathrm{t}} = \hat{\mathbf{y}}\,\frac{E_{\parallel 0}^{\mathrm{t}}}{\eta_2} e^{-jk_2 x_{\mathrm{t}}} = \hat{\mathbf{y}}\,\frac{E_{\parallel 0}^{\mathrm{t}}}{\eta_2} e^{-jk_2(x\sin\theta_{\mathrm{t}} + z\cos\theta_{\mathrm{t}})}. \tag{8.65f}$$

By matching the tangential components of $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ in both media at $z = 0$, we again obtain the relations defining Snell's laws, as well as the following expressions for the ***Fresnel reflection and transmission coefficients for parallel polarization***:

$$\Gamma_{\parallel} = \frac{E_{\parallel 0}^{\mathrm{r}}}{E_{\parallel 0}^{\mathrm{i}}} = \frac{\eta_2\cos\theta_{\mathrm{t}} - \eta_1\cos\theta_{\mathrm{i}}}{\eta_2\cos\theta_{\mathrm{t}} + \eta_1\cos\theta_{\mathrm{i}}}, \tag{8.66a}$$

$$\tau_{\parallel} = \frac{E_{\parallel 0}^{\mathrm{t}}}{E_{\parallel 0}^{\mathrm{i}}} = \frac{2\eta_2\cos\theta_{\mathrm{i}}}{\eta_2\cos\theta_{\mathrm{t}} + \eta_1\cos\theta_{\mathrm{i}}}. \tag{8.66b}$$

The preceding expressions can be shown to yield the relation

$$\tau_{\parallel} = (1 + \Gamma_{\parallel})\,\frac{\cos\theta_{\mathrm{i}}}{\cos\theta_{\mathrm{t}}}. \tag{8.67}$$

We noted earlier in connection with the perpendicular-polarization case that, when the second medium is a perfect conductor with $\eta_2 = 0$, the incident wave gets totally reflected at the boundary. The same is true for the parallel polarization case; setting $\eta_2 = 0$ in Eqs. (8.66a and b) gives $\Gamma_{\parallel} = -1$ and $\tau_{\parallel} = 0$.

For nonmagnetic materials, Eq. (8.66a) becomes

$$\Gamma_{\parallel} = \frac{-(\varepsilon_2/\varepsilon_1)\cos\theta_{\mathrm{i}} + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_{\mathrm{i}}}}{(\varepsilon_2/\varepsilon_1)\cos\theta_{\mathrm{i}} + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_{\mathrm{i}}}} \quad (\text{for } \mu_1 = \mu_2). \tag{8.68}$$

8-4.2 平行偏振

如果我们交换 E 和 H 在上一小节介绍的垂直偏振场景中所扮演的角色，同时牢记 E×H 必须指向每个入射波、反射波和透射波的传播方向的要求，我们最终会得到图 8-16 所示的平行偏振场景。现在电场位于入射平面内，而相关的磁场则垂直于入射平面。参考图 8-16 所示的方向，入射波、反射波和透射波的场由下式给出

事件波

$$\widetilde{\mathbf{E}}_{\parallel}^{\mathrm{i}} = \hat{\mathbf{y}}_{\mathrm{i}} E_{\parallel 0}^{\mathrm{i}} e^{-jk_1 x_{\mathrm{i}}} = (\hat{\mathbf{x}}\cos\theta_{\mathrm{i}} - \hat{\mathbf{z}}\sin\theta_{\mathrm{i}}) E_{\parallel 0}^{\mathrm{i}} e^{-jk_1(x\sin\theta_{\mathrm{i}} + z\cos\theta_{\mathrm{i}})}, \tag{8.65a}$$

$$\widetilde{\mathbf{H}}_{\parallel}^{\mathrm{i}} = \hat{\mathbf{y}}\,\frac{E_{\parallel 0}^{\mathrm{i}}}{\eta_1} e^{-jk_1 x_{\mathrm{i}}} = \hat{\mathbf{y}}\,\frac{E_{\parallel 0}^{\mathrm{i}}}{\eta_1} e^{-jk_1(x\sin\theta_{\mathrm{i}} + z\cos\theta_{\mathrm{i}})}, \tag{8.65b}$$

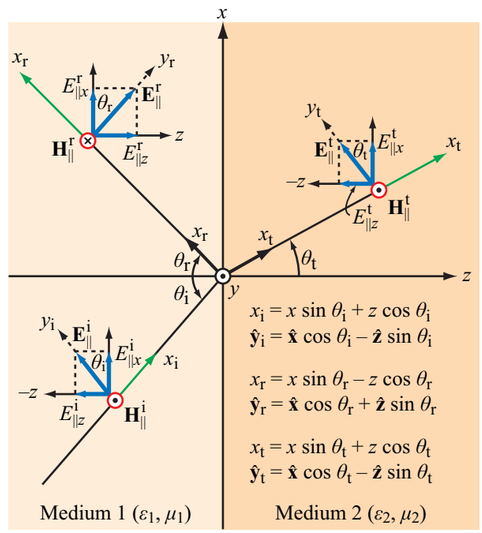


图 8-16 平行偏振平面波以角度 θ_{i} 入射到平面边界上。

反射波

$$\widetilde{\mathbf{E}}_{\parallel}^{\mathrm{r}} = \hat{\mathbf{y}}_{\mathrm{r}} E_{\parallel 0}^{\mathrm{r}} e^{-jk_1 x_{\mathrm{r}}} = (\hat{\mathbf{x}}\cos\theta_{\mathrm{r}} + \hat{\mathbf{z}}\sin\theta_{\mathrm{r}}) E_{\parallel 0}^{\mathrm{r}} e^{-jk_1(x\sin\theta_{\mathrm{r}} - z\cos\theta_{\mathrm{r}})}, \tag{8.65c}$$

$$\widetilde{\mathbf{H}}_{\parallel}^{\mathrm{r}} = -\hat{\mathbf{y}}\,\frac{E_{\parallel 0}^{\mathrm{r}}}{\eta_1} e^{-jk_1 x_{\mathrm{r}}} = -\hat{\mathbf{y}}\,\frac{E_{\parallel 0}^{\mathrm{r}}}{\eta_1} e^{-jk_1(x\sin\theta_{\mathrm{r}} - z\cos\theta_{\mathrm{r}})}, \tag{8.65d}$$

传输波

$$\widetilde{\mathbf{E}}_{\parallel}^{\mathrm{t}} = \hat{\mathbf{y}}_{\mathrm{t}} E_{\parallel 0}^{\mathrm{t}} e^{-jk_2 x_{\mathrm{t}}} = (\hat{\mathbf{x}}\cos\theta_{\mathrm{t}} - \hat{\mathbf{z}}\sin\theta_{\mathrm{t}}) E_{\parallel 0}^{\mathrm{t}} e^{-jk_2(x\sin\theta_{\mathrm{t}} + z\cos\theta_{\mathrm{t}})}, \tag{8.65e}$$

$$\widetilde{\mathbf{H}}_{\parallel}^{\mathrm{t}} = \hat{\mathbf{y}}\,\frac{E_{\parallel 0}^{\mathrm{t}}}{\eta_2} e^{-jk_2 x_{\mathrm{t}}} = \hat{\mathbf{y}}\,\frac{E_{\parallel 0}^{\mathrm{t}}}{\eta_2} e^{-jk_2(x\sin\theta_{\mathrm{t}} + z\cos\theta_{\mathrm{t}})}. \tag{8.65f}$$

通过匹配 z = 0 处两种介质中 ˜E 和 ˜H 的切向分量，我们再次获得定义斯涅尔定律的关系，以及平行偏振菲涅尔反射和透射系数的以下表达式：

$$\Gamma_{\parallel} = \frac{E_{\parallel 0}^{\mathrm{r}}}{E_{\parallel 0}^{\mathrm{i}}} = \frac{\eta_2\cos\theta_{\mathrm{t}} - \eta_1\cos\theta_{\mathrm{i}}}{\eta_2\cos\theta_{\mathrm{t}} + \eta_1\cos\theta_{\mathrm{i}}}, \tag{8.66a}$$

$$\tau_{\parallel} = \frac{E_{\parallel 0}^{\mathrm{t}}}{E_{\parallel 0}^{\mathrm{i}}} = \frac{2\eta_2\cos\theta_{\mathrm{i}}}{\eta_2\cos\theta_{\mathrm{t}} + \eta_1\cos\theta_{\mathrm{i}}}. \tag{8.66b}$$

前面的表达式可以得出以下关系

$$\tau_{\parallel} = (1 + \Gamma_{\parallel})\,\frac{\cos\theta_{\mathrm{i}}}{\cos\theta_{\mathrm{t}}}. \tag{8.67}$$

我们之前在垂直偏振情况下注意到，当第二介质是 $\eta_2 = 0$ 的理想导体时，入射波在边界处发生全反射。对于平行偏振的情况也是如此；在等式中设置 $\eta_2 = 0$ (8.66a 和 b) 给出 $\Gamma_{\parallel} = -1$ 和 $\tau_{\parallel} = 0$。

对于非磁性材料，方程。 (8.66a) 变为

$$\Gamma_{\parallel} = \frac{-(\varepsilon_2/\varepsilon_1)\cos\theta_{\mathrm{i}} + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_{\mathrm{i}}}}{(\varepsilon_2/\varepsilon_1)\cos\theta_{\mathrm{i}} + \sqrt{(\varepsilon_2/\varepsilon_1) - \sin^2\theta_{\mathrm{i}}}} \quad (\text{for } \mu_1 = \mu_2). \tag{8.68}$$

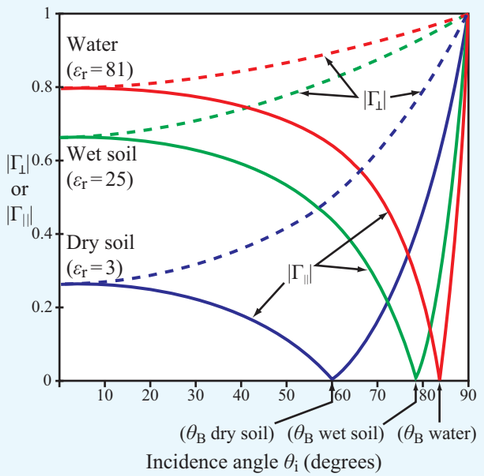


Figure 8-17 Plots for $|\Gamma_\perp|$ and $|\Gamma_\parallel|$ as a function of θ_i for a dry soil surface, a wet soil surface, and a water surface. For each surface, $|\Gamma_\parallel| = 0$ at the Brewster angle.

To illustrate the angular variations of the magnitudes of $\Gamma_\perp$ and $\Gamma_\parallel$, **Fig. 8-17** shows plots for waves incident in air onto three different types of dielectric surfaces: dry soil ($\varepsilon_r = 3$), wet soil ($\varepsilon_r = 25$), and water ($\varepsilon_r = 81$). For each of the surfaces, (1) $\Gamma_\perp = \Gamma_\parallel$ at normal incidence ($\theta_i = 0$), as expected; (2) $|\Gamma_\perp| = |\Gamma_\parallel| = 1$ at ***grazing incidence*** ($\theta_i = 90°$); and (3) $\Gamma_\parallel$ goes to zero at an angle called the ***Brewster angle*** in **Fig. 8-17**. Had the materials been magnetic too ($\mu_1 \neq \mu_2$), it would have been possible for $\Gamma_\perp$ to vanish at some angle as well. However, for nonmagnetic materials, the Brewster angle exists only for parallel polarization, and its value depends on the ratio ($\varepsilon_2/\varepsilon_1$), as we see shortly.

► At the Brewster angle, the parallel-polarized component of the incident wave is totally transmitted into medium 2. ◄

8-4.3 Brewster Angle

The Brewster angle θ_B is defined as the incidence angle θ_i at which the Fresnel reflection coefficient $\Gamma = 0$.

Perpendicular Polarization

For perpendicular polarization, the Brewster angle $\theta_{B\perp}$ can be obtained by setting the numerator of the expression for $\Gamma_\perp$, given by Eq. (8.58a), equal to zero. This happens when

$$\eta_2 \cos\theta_i = \eta_1 \cos\theta_t. \tag{8.69}$$

By (1) squaring both sides of Eq. (8.69), (2) using Eq. (8.56), (3) solving for θ_i, and then denoting θ_i as $\theta_{B\perp}$, we obtain

$$\sin\theta_{B\perp} = \sqrt{\frac{1-(\mu_1\varepsilon_2/\mu_2\varepsilon_1)}{1-(\mu_1/\mu_2)^2}}. \tag{8.70}$$

Because the denominator of Eq. (8.70) goes to zero when $\mu_1 = \mu_2$, $\theta_{B\perp}$ *does not exist for nonmagnetic materials.*

Parallel Polarization

For parallel polarization, the Brewster angle $\theta_{B\parallel}$ at which $\Gamma_\parallel = 0$ can be found by setting the numerator of $\Gamma_\parallel$ in Eq. (8.66a) equal to zero. The result is identical to Eq. (8.70), but with μ and ε interchanged. That is,

$$\sin\theta_{B\parallel} = \sqrt{\frac{1-(\varepsilon_1\mu_2/\varepsilon_2\mu_1)}{1-(\varepsilon_1/\varepsilon_2)^2}}. \tag{8.71}$$

For nonmagnetic materials,

$$\theta_{B\parallel} = \sin^{-1}\sqrt{\frac{1}{1+(\varepsilon_1/\varepsilon_2)}} = \tan^{-1}\sqrt{\frac{\varepsilon_2}{\varepsilon_1}} \qquad (\text{for } \mu_1 = \mu_2). \tag{8.72}$$

The Brewster angle is also called the ***polarizing angle***. This is because, if a wave composed of both perpendicular and parallel polarization components is incident upon a nonmagnetic surface at the Brewster angle $\theta_{B\parallel}$, the parallel polarized component is totally transmitted into the second medium, and only the perpendicularly polarized component is reflected by the surface. Natural light, including sunlight and light generated by most manufactured sources, is ***unpolarized*** because the direction of the electric field of the light waves varies randomly in angle over the plane perpendicular to the direction of propagation. Thus, on average, half of the intensity of natural light is perpendicularly polarized and the other half is parallel polarized. When unpolarized light is incident upon a surface at the Brewster angle, the reflected wave is strictly perpendicularly polarized. Hence, the surface acts as a polarizer.

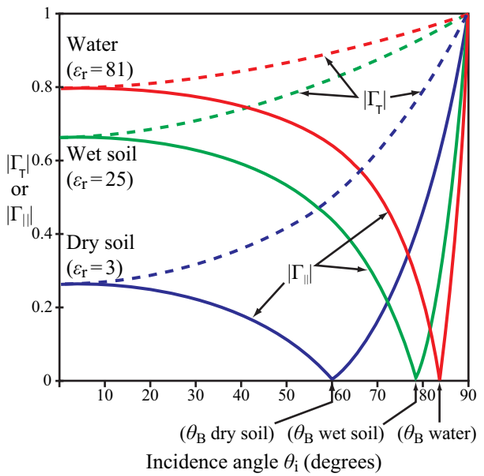


图 8-17 针对干燥土壤表面、湿土壤表面和水面，将 $|\Gamma_\perp|$ 和 $|\Gamma_\parallel|$ 作为 θ_i 的函数绘制。对于每个表面，布儒斯特角处的 $|\Gamma_\parallel| = 0$。

为了说明 $\Gamma_\perp$ 和 $\Gamma_\parallel$ 幅度的角度变化，图 8-17 显示了空气中入射到三种不同类型介电表面的波的绘图：干燥土壤 ($\varepsilon_r = 3$)、湿土壤 ($\varepsilon_r = 25$) 和水 ($\varepsilon_r = 81$)。对于每个表面，(1) 垂直入射时的 $\Gamma_\perp = \Gamma_\parallel$ ($\theta_i = 0$)，如预期； (2) 掠射入射角处的 $|\Gamma_\perp| = |\Gamma_\parallel| = 1$ ($\theta_i = 90°$)； (3) $\Gamma_\parallel$ 在图 8-17 中称为布儒斯特角的角度处为零。如果这些材料也具有磁性 ($\mu_1 \neq \mu_2$)，那么 $\Gamma_\perp$ 也有可能在某个角度消失。然而，对于非磁性材料，布儒斯特角仅存在于平行极化时，其值取决于比率 ($\varepsilon_2/\varepsilon_1$)，正如我们稍后看到的。

► 在布儒斯特角处，入射波的平行偏振分量完全传输到介质 2 中。◄

8-4.3 布鲁斯特角

布鲁斯特角 θ_B 定义为菲涅耳反射系数 $\Gamma =$ 为 0 时的入射角 θ_i。

垂直偏振

对于垂直偏振，布儒斯特角 $\theta_{B\perp}$ 可以通过设置 $\Gamma_\perp$ 表达式的分子来获得，如式 (1) 所示。(8.58a)，等于零。发生这种情况时

$$\eta_2 \cos\theta_i = \eta_1 \cos\theta_t. \tag{8.69}$$

通过 (1) 对方程两边求平方 (8.69)， (2) 使用方程。(8.56), (3) 求解θ_i，然后将θ_i表示为$\theta_{B\perp}$，我们得到

$$\sin\theta_{B\perp} = \sqrt{\frac{1-(\mu_1\varepsilon_2/\mu_2\varepsilon_1)}{1-(\mu_1/\mu_2)^2}}. \tag{8.70}$$

因为方程的分母。当非磁性材料不存在 $\mu_1 = \mu_2$、$\theta_{B\perp}$ 时，(8.70) 为零。

平行偏振

对于平行偏振，可以通过设置式 (1) 中$\Gamma_\parallel$的分子来找到 $\Gamma_\parallel$ =为0时的布儒斯特角$\theta_{B\parallel}$。(8.66a) 等于零。结果与等式相同。 (8.70)，但 μ 和 ε 互换。那是，

$$\sin\theta_{B\parallel} = \sqrt{\frac{1-(\varepsilon_1\mu_2/\varepsilon_2\mu_1)}{1-(\varepsilon_1/\varepsilon_2)^2}}. \tag{8.71}$$

对于非磁性材料，

$$\theta_{B\parallel} = \sin^{-1}\sqrt{\frac{1}{1+(\varepsilon_1/\varepsilon_2)}} = \tan^{-1}\sqrt{\frac{\varepsilon_2}{\varepsilon_1}} \qquad (\text{for } \mu_1 = \mu_2). \tag{8.72}$$

布鲁斯特角也称为偏振角。这是因为，如果由垂直和平行偏振分量组成的波以布儒斯特角 $\theta_{B\parallel}$ 入射到非磁性表面，则平行偏振分量将完全透射到第二介质中，而只有垂直偏振分量被表面反射。自然光，包括太阳光和大多数人造光源产生的光，都是非偏振的，因为光波的电场方向在垂直于传播方向的平面上的角度随机变化。因此，平均而言，自然光强度的一半是垂直偏振的，另一半是平行偏振的。当非偏振光以布儒斯特角入射到表面时，反射波是严格垂直偏振的。因此，该表面充当偏振器。